

Temporal Considerations in Clinical Machine Learning: A Diagnostic Framework for Prospective and Local Validation

Corresponding Author: Professor Tina Hernandez-Boussard

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this analysis, the authors present model-agnostic diagnostic framework to validate machine learning models and show utility on a cancer acute care prediction model. Their framework detects temporal shifts in model covariates and outcomes, identifying real-time (or at least year to year) tradeoffs in retraining timing and efficacy.

I have several comments:

1) I am concerned about the strategy for missing data characterization described in lines 120-121 for the model. Missingness of a data element may be a common contributor to drift (e.g. if a laboratory becomes renamed, if more patients stop showing up to the clinic during COVID), but an imputation-based approach using sample mean likely masks this contributors to drift. Can the authors comment on their rationale for this approach and explore more methods (e.g. missing indicator, etc.) for accounting for variable missingness?

2) While I agree with the feature reduction framework proposed by the authors, I wonder if the authors can also simulate including more variables that “come online” – AKA adding features as they become available temporally. This is relevant because generally when retraining EHR models in practice, we would use whichever variables are available in the EHR prior to feature reduction. I am unsure if the authors are only removing “noisy” features over time in their framework, which may lead to worse performance over time.

3) Re: permutation importance: I struggle with how this method accounts for highly correlated features or features that may “drift together” in subtle ways. I worry about removing features when other correlated features are left, or unintentionally dropping features that are truly causal from the model. Can you comment on how robust this strategy is to highly correlated features?

4) One should note that the number of features (up to 1000) is quite high relative to the the number of observations and cases in both the training and validation datasets, leading to some concern over overfitting

Reviewer #2

(Remarks to the Author)

This is a thoughtful, well structured analysis for temporal (drift) effects on ml models.

Overall, the framework is strong and well thought out. However, it is difficult to understand the specific impact of each step. The authors should clarify the success criteria for each of the 4 steps, and highlight how their approach helps understand the temporal impact.

Older and newer data was not as powerful. The discussion lacks clarity for why that was.

More specific comments on this space would be helpful. In oncology, how did your model need to be assisted?

Reviewer #3

(Remarks to the Author)

The manuscript entitled "Temporal Considerations in Clinical Machine Learning: A Diagnostic Framework for Prospective and Local Validation" presents an extremely interesting and timely investigation about the variability issue in learning

frameworks. In particular, the authors use explainability to ensure a greater robustness and interpretability to their analyses. The work is well written, clear and pleasant to read. My major concern, but I would probably consider it a further improvement of an already excellent work, is that the authors examine only one, although formidably extended, case study: acute care utilization. I think it will be of great interest to see how the framework applies to other case studies and to which extent the conclusions can be generalized. In any case, I recommend this work for publication.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This is an excellent response and I have no further concerns

Reviewer #3

(Remarks to the Author)

The authors addressed my comment by suggesting further possible case studies.

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

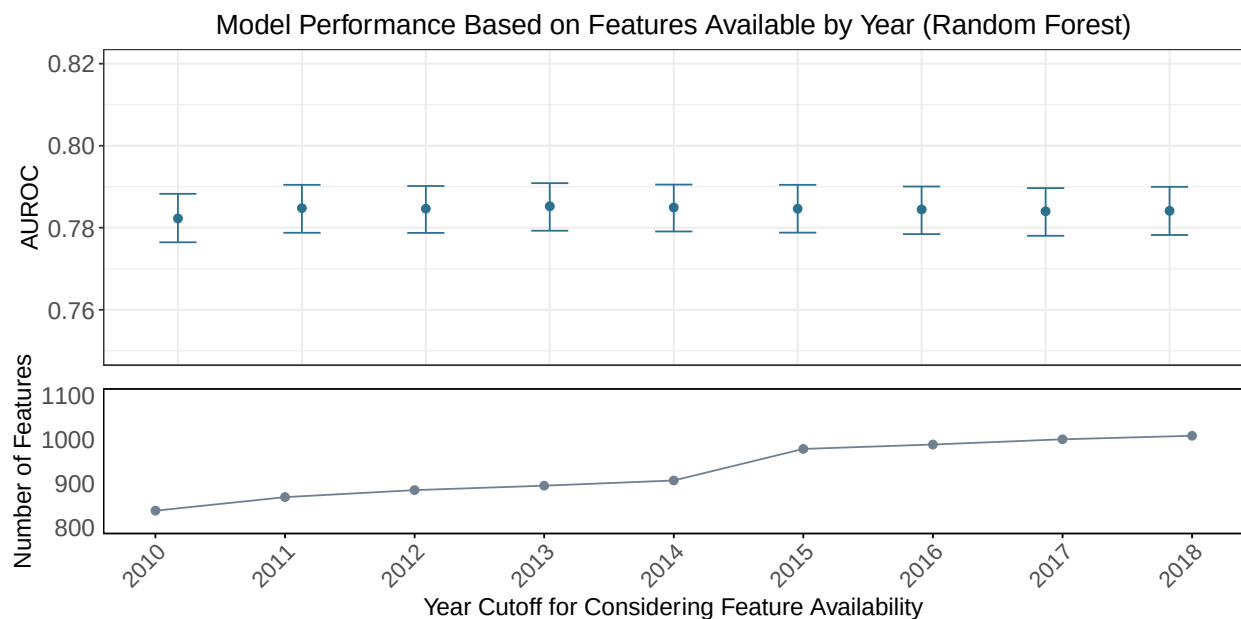
1) I am concerned about the strategy for missing data characterization described in lines 120-121 for the model. Missingness of a data element may be a common contributor to drift (e.g. if a laboratory becomes renamed, if more patients stop showing up to the clinic during COVID), but an imputation-based approach using sample mean likely masks this contributors to drift. Can the authors comment on their rationale for this approach and explore more methods (e.g. missing indicator, etc.) for accounting for variable missingness?

We agree with the Reviewer on this important point. In our study we used Last Observation Carried Forward (LOCF) and mean-based imputation approaches because both are easy to understand, conservative, and not dependent on hyperparameter-tuning (and therefore generalizable across applications). In our revision, we implement k -nearest neighbor (KNN)-based imputation. This approach has a few advantages. First, KNN-based imputation is a well-established, highly effective strategy in many settings. Moreover, by tuning k , our imputed values become more or less susceptible to drift. For example, if drift is strong but gradual, setting k to smaller values will allow the KNN imputation to rely primarily on the closest (and potentially most recent) neighbors. By increasing k , an increasing number of neighbors from more distant years is used in the imputation, allowing the imputation to be based on more observations. In contrast, if the data were affected by periodic, sudden drift, choosing a larger k might help offset shocks. To highlight how the hyperparameter k affects performance, we implemented this KNN-based imputation approach for $k = 5, 15, 100$ and 1000 . We also agree with you that missingness indicators can be a powerful strategy. In our case, this option appeared less feasible because it would have doubled the number of diagnosis and procedure code features (>500 variables), resulting in more dimensions and, thus, risk for overfitting. Given the importance of imputation in settings with drift, we have added this point to our discussion and thank the Reviewer for bringing this point to our attention (l. 375ff).

2) While I agree with the feature reduction framework proposed by the authors, I wonder if the authors can also simulate including more variables that “come online” – AKA adding features as they become available temporally. This is relevant because generally when retraining EHR models in practice, we would use whichever variables are available in the EHR prior to feature reduction. I am unsure if the authors are only removing “noisy” features over time in their framework, which may lead to worse performance over time.

We agree with you that this is an excellent experiment that highlights how the number of features changes over time. As a result, we designed an experiment in which features we show model performance as features are progressively added to the model. In our plot below we show the results from this experiment for our candidate model (Random Forest). The x-axis denotes the year cutoff up to which features that are already available are included. The y-axis shows the performance on the held-out prospective test set as features come online. The lower panel displays the number of features included in the

model, i.e. features available in this year. Since there is no clear performance impact when adding newer features to the model, the differential performance across years appears to be driven by features that are already included in 2010 ($p_{features} = 837$). Despite this “null result”, we believe this experiment could potentially be powerful in other applications and also highlights the dynamic of the feature set over time. As a result, we have added this to the supplement (S5).



3) Re: permutation importance: I struggle with how this method accounts for highly correlated features or features that may “drift together” in subtle ways. I worry about removing features when other correlated features are left, or unintentionally dropping features that are truly causal from the model. Can you comment on how robust this strategy is to highly correlated features?

We thank Reviewer 1 for their excellent question on permutation importance (PI). We chose this feature importance metric for two reasons. First, PI is a model-agnostic feature importance metric and therefore generalizable across models. Since we deliberately designed our framework to be data-centric and broadly applicable, choosing a model-agnostic performance metric was a key priority. Second, this approach also ensures that features are evaluated for their predictive performance, whereas SHAP values test how much a specific feature contributes to the prediction. As you point out, if a feature is important, but highly correlated, its shuffling might not necessarily result in a performance drop. To test this hypothesis, we designed a simulation study with $2 * p$ covariates, consisting of p pairs of perfectly correlated ($\rho = 1.0$) features. We then removed one feature at a time from these $2 * p$ covariates recursively until only 5 features were left. Figure R.1 displays the results from this recursive feature elimination experiment based on PI for $p = 50$ and sample size $n = 2000$. The blue curve depicts performance (AUROC,

left y-axis). The red curve shows the number of feature pairs removed (right y-axis) as features are iteratively removed (x-axis). As feature pairs begin to get removed early in the process, the red curve declines well before 50 features. If permutation importance resulted in scores of 0 under strong correlation, we would expect the red curve to remain flat until the number of total features remaining reached 50 (all identical copies are removed). The early decline of the red curve illustrates that correlated features get removed much earlier in this process, suggesting that correlation might not lead to removal of important features but rather create some delay in feature removal.

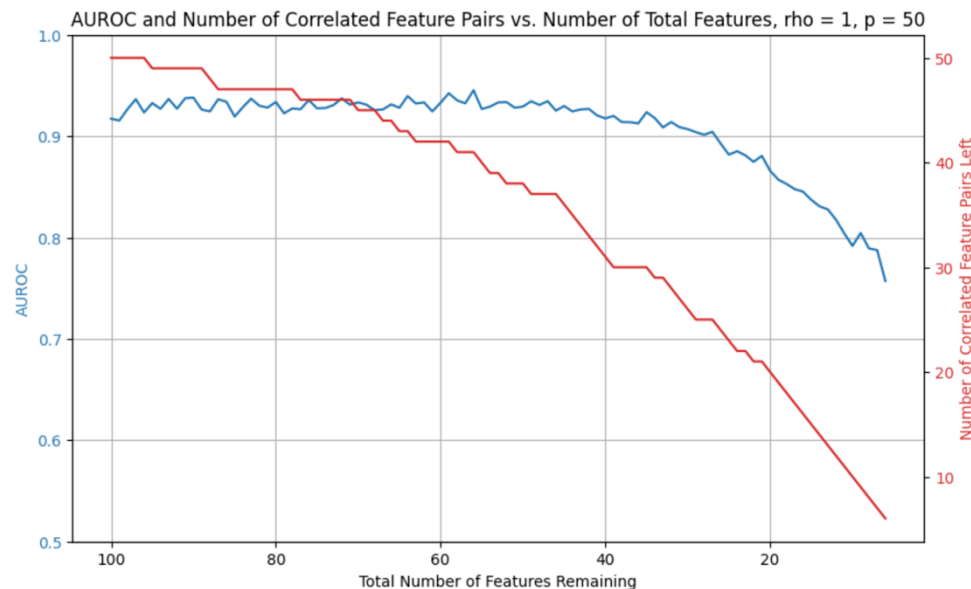
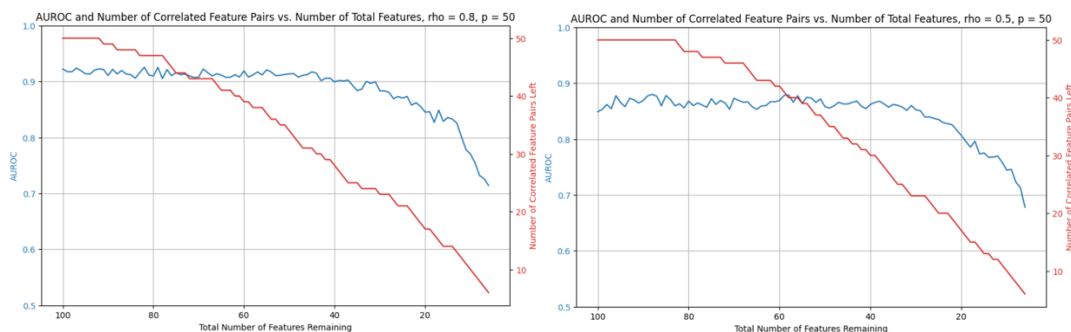


Figure R.1 The plot illustrates the evolution of performance (blue curve) and number of correlated features remaining (red curve) in the feature set when using permutation importance for recursive feature elimination. The x-axis indicates the total number of features remaining in the feature set. The experiment is conducted with 50 pairs of perfectly correlated features, resulting in a total of 100 features.

We also tested this approach for different correlation coefficients of $\rho = 0.5$ and 0.8 (see results below).



Since high feature correlation might be a concern for some users, we have added a caveat in our discussion (l.314ff), suggesting that users might consider using hierarchical clustering based on the Spearman rank-order correlations as recommended in the literature.[1] We also agree with the reviewer that drift can be an important concern in this context. To address this issue, we only use data from future test years to measure the importance of each feature. Features under strong drift are less predictive and thus more likely to be removed under this approach.

[1] https://scikit-learn.org/stable/auto_examples/inspection/plot_permutation_importance_multicollinear.html

4) One should note that the number of features (up to 1000) is quite high relative to the the number of observations and cases in both the training and validation datasets, leading to some concern over overfitting

We thank Reviewer 1 for this excellent comment and agree that overfitting is an important concern, especially as electronic health records are becoming increasingly high-dimensional. For this reason, we designed experiment 1 s.t. overfitting can be tested on independent test sets both from the same time period and from future time periods. To show that we are not overfitting, we are also testing on held-out data from the same period as the train data. Furthermore, we tune hyperparameters on a held-out development set and evaluate model performance on a test set that has never been seen by the model. So as shown in our illustration, we have implemented a very strict approach that mitigates overfitting and ensures that there is no leakage between training and test data for every experiment. Please also note that we deliberately decided to include as many features as possible, including detailed histology and treatment information. Despite including many covariates, the number of samples (n) is still larger than the number of features (p), i.e. we do not have setting $p \gg n$ that necessitates regularization. The results also show that our models generalize well. Finally, the models used in this study (e.g. lasso and random forest), have regularization properties, which could address this concern even in settings with $p \gg n$.

Reviewer #2 (Remarks to the Author):

Overall, the framework is strong and well thought out. However, it is difficult to understand the specific impact of each step. The authors should clarify the success criteria for each of the 4 steps, and highlight how their approach helps understand the temporal impact.

Older and newer data was not as powerful. The discussion lacks clarity for why that was. More specific comments on this space would be helpful. In oncology, how did your model need to be assisted?

We thank the Reviewers for their excellent comment on the success criteria at each step. To improve the clarity in our framework, we have added “red flags” to Figure 1. These red flags highlight potential risks/failure modes when training models on data from extended time periods. As a result, the absence of red flags can be interpreted as a success metric.

In the presence of a red flag, researchers should avoid training the model 'off the shelf.' Instead, they should critically assess their data selection and model training process. For example, they might consider using only recent data, as integrating older data could negatively impact model performance.

We further thank the Reviewer for the point raised as to why older and newer data was not as powerful. This is an excellent point of discussion since the degree of drift varies by application. In our data, the baseline level of temporal drift is moderate, which may explain why using older training years still yields reasonable performance. This may be partly due to the presence of many “stable features” as illustrated in the heatmap. These features remain consistent over time, thereby mitigating the impact of drift on model performance. However, sudden external shocks such as the COVID-19 pandemic introduce substantial and rapid changes to the data, potentially leading to increased drift with the newer data. This could offset the relative benefits of more recent data, as these external factors introduce new variability. As a result, temporal drift might be more visible in the absence of such external shocks. We also added this interpretation to our discussion section (l.389ff).

Finally, the question as to whether the model needs to be assisted offers a great additional point for discussion that we have integrated into our discussion (l. 345ff). We have added this point to our discussion section, since the distinction between data- and model-centric approach is highly relevant and has gained much attention in recent years. In this study, we adopt a model-agnostic and data-centric view, as suggested by Andrew Ng and others. Specifically, we focus on the assessment of data quality, quantity, and relevance prior to model training. Rather than assisting a specific model, we focus on the data and allow researchers to apply a model of their choosing within the framework. This approach has two advantages. First, it aligns with the idea that in the presence of noisy or low-quality data sophisticated models often perform poorly. This view is corroborated by research that highlights the susceptibility of complex models to noisy data environments; a focus on data curation and simple models, therefore, often turns out to be a better approach in real-world environments (Northcutt et al., NeurIPS, 2021). A data-centric view also aligns with the responsible, fair and interpretable use of AI since data exploration and curation provides guarantees for the reliability and interpretability of model outputs. Second, this data-centric approach ensures the generalizability of the framework across model architectures. Since the requirement for different model architectures can differ from one application to another, a model agnostic framework is likely to stay relevant over time and across fields.

Reviewer #3 (Remarks to the Author):

The manuscript entitled "Temporal Considerations in Clinical Machine Learning: A Diagnostic Framework for Prospective and Local Validation" presents an extremely interesting and timely investigation about the variability issue in learning frameworks. In particular, the authors use explainability to ensure a greater robustness and interpretability to their analyses. The work is well written, clear and pleasant to read.

My major concern, but I would probably consider it a further improvement of an already excellent work, is that the authors examine only one, although formidably extended, case study: acute care utilization. I think it will be of great interest to see how the framework applies to other case studies and to which extent the conclusions can be generalized. In any case, I recommend this work for publication.

We thank Reviewer 3 for their great comments. As you rightly point out, the generalizability of machine learning frameworks is crucial for their impact. Generalizability can be demonstrated through either a broad array of applications, or by focusing on a highly complex and multifaceted application that encompasses various challenges and variables, allowing for easier adaptation to less complex scenarios. In our study, we chose to prioritize depth over breadth by selecting a single yet highly complex application. This approach was intentional to ensure that our framework is robust and applicable to a wide range of applications. Furthermore, this is complex as it integrates multiple outcomes (ED visits and hospital admissions), tumor types (and thus diseases), disease stages, and data types (e.g. ranging from demographic data to treatment-specific information) across a relatively long time span of 13 years. In addition, our cohort is likely to be affected by both sudden drift, such as the COVID-19 pandemic, and more gradual temporal drift, further increasing the complexity. This complexity allows us to showcase the robustness and full utility of the framework. To further illustrate the potential for generalizability, we have added future directions for research in the limitations/future research section of our discussion (see final paragraph in discussion section, l.433ff). For example, conditions such as essential arterial hypertension, where guidelines have evolved over time, could serve as an interesting use case for our framework.

Rebuttals – Letter to the editor

We thank the editors and reviewers for their valuable feedback and comments. In response to the final review of our manuscript, we have carefully followed the editorial checklist and made the following modifications.

The title has been changed to “Diagnostic framework to validate clinical machine learning models locally on temporally stamped data”. We agree that this title reflects the content and purpose of our study well.

We further shortened and restructured the abstract to comply with the format of a structured abstract. We provide a plain summary statement that explains our study to a wider audience. To align with the editorial guidelines, we further changed the tense from past to present in the results and discussion sections. We relabeled the figures to provide figure labels with lower capital letters. We replaced Figure 6b by a boxplot to show the distribution of points more effectively. We also phrased sentences containing language such as “novel”, “state-of-the-art” in a more neutral way and removed qualifying adjectives, where appropriate. We removed the word “significantly” and “significant” in the results section.

We provide two summary sentences: “Schuessler et al. present a multi-staged diagnostic framework for the local and temporal validation of clinical machine learning models. They illustrate the framework’s application, encompassing model performance evaluation, the temporal evolution of features and outcomes, training schedules, and strategies for data reduction and valuation, to predict acute care utilization in cancer patients.

Finally, we attach a featured image (corresponding to Figure 1 in the main paper). We also upload all images separately and provide the figures in the supplement section in a single file. We attach the final review checklist detailing the revision instructions to this rebuttal letter with answers for each item.

Thank you for considering our work for publication in Communications Medicine. We are grateful for your consideration and look forward to your reply.